Let $\mathcal{A}_p = \{\underline{x} : r_p(\underline{x}) > 1\} = \{\underline{x} : \|\underline{x}\|_p > 1\}$

The $r_p$-Pareto process $Z^{(r_p)}$ has distribution:

$$\mathbb{P}\{Z^{(r_p)} \in \cdot\} = \frac{\nu(\cdot \cap \mathcal{A}_p)}{\nu(\mathcal{A}_p)} = \frac{\nu(\cdot \cap \mathcal{A}_p)}{\theta_p}$$

where $\theta_p = \nu(\mathcal{A}_p)$
$= \nu\{\underline{x} : r_p(\underline{x}) > 1\}$

Now assume another risk function $r_0$ s.t $\boxed{r_0 \le \Pi_p r_p}$

Take $r_0(\underline{x}) = \|\underline{x}\|_1$, by Hölder inequality $\|\underline{x}\|_1 \le D^{1-1/p} \|\underline{x}\|_p$

$\boxed{\text{The } r_p\text{-Pareto draw is accepted whenever } r_0(Z^{(r_p)}) > q}$, $q > \Pi_p$

- $\mathcal{A}_1 \subset \frac{1}{\Pi_p}\mathcal{A}_p$ : $\underline{x} \in \mathcal{A}_1 \Rightarrow \|\underline{x}\|_1 > 1$

$$\Rightarrow \Pi_p \|\underline{x}\|_p > 1$$
$$\Rightarrow \|\Pi_p \underline{x}\|_p > 1$$
$$\Rightarrow \Pi_p \underline{x} \in \mathcal{A}_p$$
$$\Rightarrow \underline{x} \in \frac{1}{\Pi_p}\mathcal{A}_p$$

- $$\frac{\nu(B \cap \frac{1}{\Pi_p}\mathcal{A}_p)}{\nu(\frac{1}{\Pi_p}\mathcal{A}_p)} = \frac{\Pi_p^{-\alpha}\, \nu(\Pi_p B \cap \mathcal{A}_p)}{\Pi_p^{-\alpha}\, \nu(\mathcal{A}_p)}$$

$$= \frac{\nu(\Pi_p B \cap \mathcal{A}_p)}{\nu(\mathcal{A}_p)}$$

$$= \mathbb{P}\{Z^{(r_p)} \in \Pi_p B\} = \mathbb{P}\{\tfrac{1}{\Pi_p} Z^{(r_p)} \in B\}$$

- $$\mathbb{P}\{\tfrac{1}{\Pi_p} Z^{(r_p)} \in B \mid \tfrac{1}{\Pi_p} Z^{(r_p)} \in \mathcal{A}_1\} = \frac{\mathbb{P}\{\frac{1}{\Pi_p} Z^{(r_p)} \in B \cap \mathcal{A}_1\}}{\mathbb{P}\{\frac{1}{\Pi_p} Z^{(r_p)} \in \mathcal{A}_1\}}$$

$$= \frac{\nu(B \cap \mathcal{A}_1 \cap \frac{1}{\Pi_p}\mathcal{A}_p)}{\nu(\frac{1}{\Pi_p}\mathcal{A}_p)} \times \frac{\nu(\frac{1}{\Pi_p}\mathcal{A}_p)}{\nu(\mathcal{A}_1 \cap \frac{1}{\Pi_p}\mathcal{A}_p)} = \frac{\nu(B \cap \mathcal{A}_1)}{\nu(\mathcal{A}_1)} = \mathbb{P}\{Z^{(r_1)} \in B\}$$

So what's $\mathbb{P}\{ r_0 (Z^{(r_p)}) > q \}$ ?

Let $B_q = \{ \underline{x} : r_0(\underline{x}) > q \}$ , then we want $\mathbb{P}\{ Z^{(r_p)} \in B_q \}$

By definition we have $\mathbb{P}\{ Z^{(r_p)} \in B_q \} = \dfrac{\nu( B_q \cap A_p)}{\nu(A_p)}$

$$B_q \cap A_p = \{ \underline{x} : r_0(\underline{x}) > q \} \cap \{ \underline{x} : r_p(\underline{x}) > 1 \}$$
$$= \{ \underline{x} : r_0(\underline{x}) > q ,\ r_p(\underline{x}) > 1 \}$$

given that $r_0 \leq \Pi_p r_p$ ~~and $q > \Pi_p$ then we have~~

~~$(\Rightarrow r_0(\underline{x}) > \Pi_p)$~~

$r_0 > q$

then $q \leq \Pi_p r_p \Rightarrow r_p \geq \dfrac{q}{\Pi_p} > 1$ $\Rightarrow \{ \underline{x} : r_0 > q \} \subset \{ \underline{x} : r_p > 1 \}$

when $q \geq \Pi_p$

therefore

$$B_q \cap A_p = \{ \underline{x} : r_0(\underline{x}) > q ,\ r_p(\underline{x}) > 1 \}$$
$$= \{ \underline{x} : r_0(\underline{x}) > q \}$$

$$\Rightarrow \mathbb{P}\{ Z^{(r_p)} \in B_q \} = \frac{\nu(\{ \underline{x} : r_0(\underline{x}) > q \})}{\nu(\{ \underline{x} : r_p(\underline{x}) > 1 \})}$$

$B_q = \{ \underline{x} : r_0(\underline{x}) > q \} = q \{ \underline{x} : r_0(\underline{x}) > 1 \}$
$= q\, B_1$

$$= q^{-\alpha} \underbrace{\frac{\nu(\{ \underline{x} : r_0(\underline{x}) > 1 \})}{\nu(\{ \underline{x} : r_p(\underline{x}) > 1 \})}}_{\text{ratio of the normalising constants}}$$

depends on dependence structure through $\nu(\{ \underline{x} : r_p(\underline{x}) > 1 \})$

Acceptance rate is $$q^{-\alpha} \frac{\nu(\{\underline{x}: r_0(\underline{x}) > 1\})}{\nu(\{\underline{x}: r_p(\underline{x}) > 1\})}$$

For $r_0(\underline{x}) = \|\underline{x}\|_1$ and $r_p(\underline{x}) = \|\underline{x}\|_p$ , $\Pi_p = D^{1-1/p}$

So we need $q \geqslant D^{1-1/p}$

Since acceptance rate is decreasing in $q$ ($\alpha > 0$) then we should select $q = D^{1-1/p}$ to select the most observations. $\Rightarrow D^{(1/p-1)\alpha} \dfrac{\nu(\{\underline{x}: r_0(\underline{x}) > 1\})}{\nu(\{\underline{x}: r_p(\underline{x}) > 1\})}$

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Consider unit-Pareto margins ($\alpha = 1$)

write $s = \|\underline{x}\|_1$ and $\underline{x} = s\underline{w}$ , $\underline{w} \in S_1 = \{\underline{w} > \underline{0} : \sum w_j = 1\}$

the exponent measure can be decomposed

$$\nu(ds, d\underline{w}) = s^{-2}\, ds\, H(d\underline{w})$$

Given the margins are unit pareto : $\nu(\{x_j > 1\}) = 1$ $\quad j = 1, \dots, D$

$$x_j > 1 \Rightarrow s w_j > 1$$
$$\Rightarrow s > 1/w_j$$

From $\nu(\{x_j > 1\}) = 1 \Rightarrow \displaystyle\int_{S_1} \int_{1/w_j}^{\infty} s^{2}\, ds\, H(d\underline{w}) = 1$

$$\Rightarrow \int_{S_1} w_j \, H(d\underline{w}) = 1. \quad \text{for } j = 1, \dots, D$$

④ Now we look at $\nu(\{\underline{x}: r_0(\underline{x}) > 1)$

$$\nu(\{\underline{x}: r_0(\underline{x}) > 1\}) = \int_{S_1} \int_{1}^{\infty} s^{-2}\, ds\, H(d\underline{w}) = \int_{S_1} H(d\underline{w})$$

(Total mass of $H$, angular measure)

$$= \int_{S_1} \left(\sum w_j\right) H(dw)$$

$$= \sum_{j=1}^{D} \int_{S_1} w_j \, H(dw) = D$$

$r_p(\underline{x}) = \|\underline{x}\|_p = \|s\underline{w}\|_p = s\|w\|_p$

$r_p(\underline{x}) > 1 \Rightarrow s\|w\|_p > 1 \Rightarrow s > 1/\|w\|_p$

## Ⓑ Look at $\nu(\{\underline{x}: r_p(\underline{x}) > 1\})$

$$\nu(\{\underline{x}: r_p(\underline{x}) > 1\}) = \int_{S_1}\int_{1/\|w\|_p}^{\infty} s^{-2}\, ds\, H(dw)$$

$$= \int_{S_1} \|w\|_p \, H(dw)$$

$\|\underline{w}\|_1 = 1$  $\quad \|\underline{w}\|_p \leq \|\underline{w}\|_1 \leqslant D^{1-1/p}\|\underline{w}\|_p$

$$\Rightarrow \|\underline{w}\|_p < 1 < D^{1-1/p}\|\underline{w}\|_p$$

$$\Rightarrow \boxed{D^{1/p-1} < \|\underline{w}\|_p < 1}$$

$$\Rightarrow \int_{S_1} D^{1/p-1} H(dw) < \int_{S_1}\|w\|_p\, H(dw) < \int_{S_1} H(d\underline{w}) = D$$

$$\|$$

$$D^{1/p-1} \times D = D^{1/p}$$

$$\Rightarrow \boxed{D^{1/p} \leq \nu(\{\underline{x}: r_p(\underline{x}) > 1\}) < D.}$$

complete dependence. $\quad\uparrow$ asymptotic independence

From Ⓐ and Ⓑ: acceptance rate is given by

$$D^{(1/p-1)} \underbrace{\frac{\nu(\{\underline{x}: r_1(\underline{x}) > 1\})}{\nu(\{\underline{x}: r_p(\underline{x}) > 1\})}}_{\in [1; \ D^{1-1/p}]} \in [D^{1/p-1}, 1]$$

$\uparrow$ avg indep $\quad \uparrow$ complete dep.